Study of water stress in lavender using deep learning

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1 General context

This project, developed in collaboration with Florence Nicolè, an ecology researcher at the Plant Biotechnology Laboratory (LBV-pam) at Jean Monnet University, focuses on studying water stress in different varieties of lavender. The aim is to see whether it is possible to observe **morphological changes** in the plant (stems, leaves, etc.) when it is subjected to water stress, i.e., a lack of water.

In particular, the researchers would like to see if there are **early markers** of this stress that could be used to detect stress without the need for chemical analysis.

2 Methodology

2.1 Data collected

Sixty lavender plants of different varieties are studied for one month (from June 18 to July 16). They are subjected to three types of water stress: S50, S25, and S10, with S50 being the lowest and S10 the highest. The stress was applied from T0 (June 18) and the plants were observed at T2 (July 2) and T4 (July 16).

At each time point, all plants were photographed individually under similar conditions (see images in the database and in the figure). It should also be noted that some plants were taken for analysis and were therefore no longer present at subsequent time points. The table shows the number of plants available at each time point for each stress level.

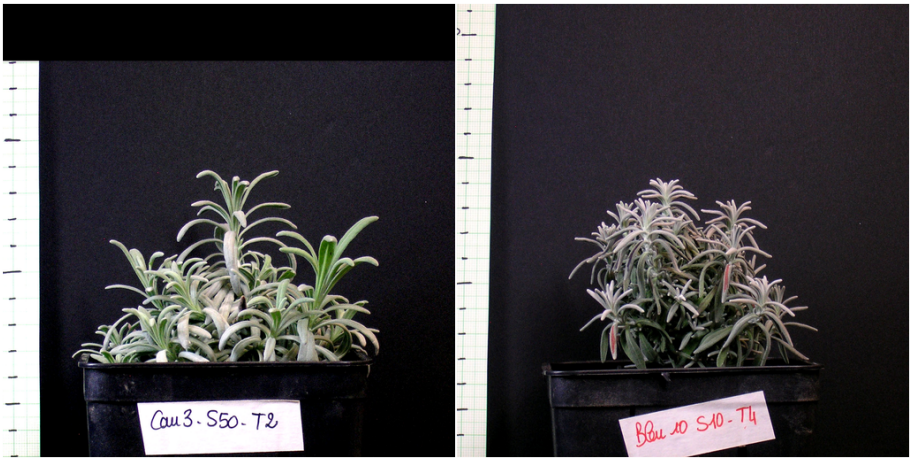

Figure 1: Examples of plants **photographed at T2**. Cau3 underwent S50 stress and was sampled at T2 for analysis, Bleu10 underwent S10 stress and was sampled at T4 for analysis.

	T0 June 18th	**T2** July 2nd	**T4** July 16th
S50	26	18	11
S25	15	15	9
S10	19	19	12
Total	60	52	32

Table 1: Number of available plants for each time and stress level.

Note that there are several varieties of lavender, which are indicated by the prefix in the images. For example, the prefix `bleu` corresponds to blue lavender, the prefix `sara` corresponds to the *Sara* variety of lavender, etc. The different varieties have different shapes and leaves.

2.2 Approach

The principle is to formulate the problem as an image classification problem for a given time, to determine whether it is possible to learn to recognize a plant that has undergone a high level of stress compared to a plant that has undergone a low level of stress. For example, at $T4$, we can learn a classifier that can differentiate between the three stress levels S50, S25, and S10. If the classifier performs well, it means that we can detect the stress level by looking at the appearance of the plant.

In a second step, to identify the elements of the image that made it possible to recognize the stress level, an activation map (such as *grad-CAM*) can be used.

Another approach could be to cut the images into small areas to examine smaller details.

The architecture of the chosen network(s) is left to your discretion.

3 Image dataset

The archive provided contains three folders, `T0`, `T2`, and `T4`, containing the plants photographed at times T0, T2, and T4, respectively. It can be downloaded here.

The name of each image is in the following format: `varxx_Syy_Tz.png` where:

- `var` represents the name of the variety: `bleu`, `cau`, `sara`, or `tel`,
- `xx` represents the plant number in the variety in question,
- `yy` represents the stress level: 50, 25, or 10,
- `z` represents the time at which the plant was sampled (note that this is not the time at which it was photographed).

For example, the photo `bleu10_S10_T4.png` in the `T2` folder corresponds to the photo at `T2` of plant number 10 of the `bleu` variety that underwent stress S10. This plant was sampled at `T4` for analysis (this last piece of information will not be used in the analysis).